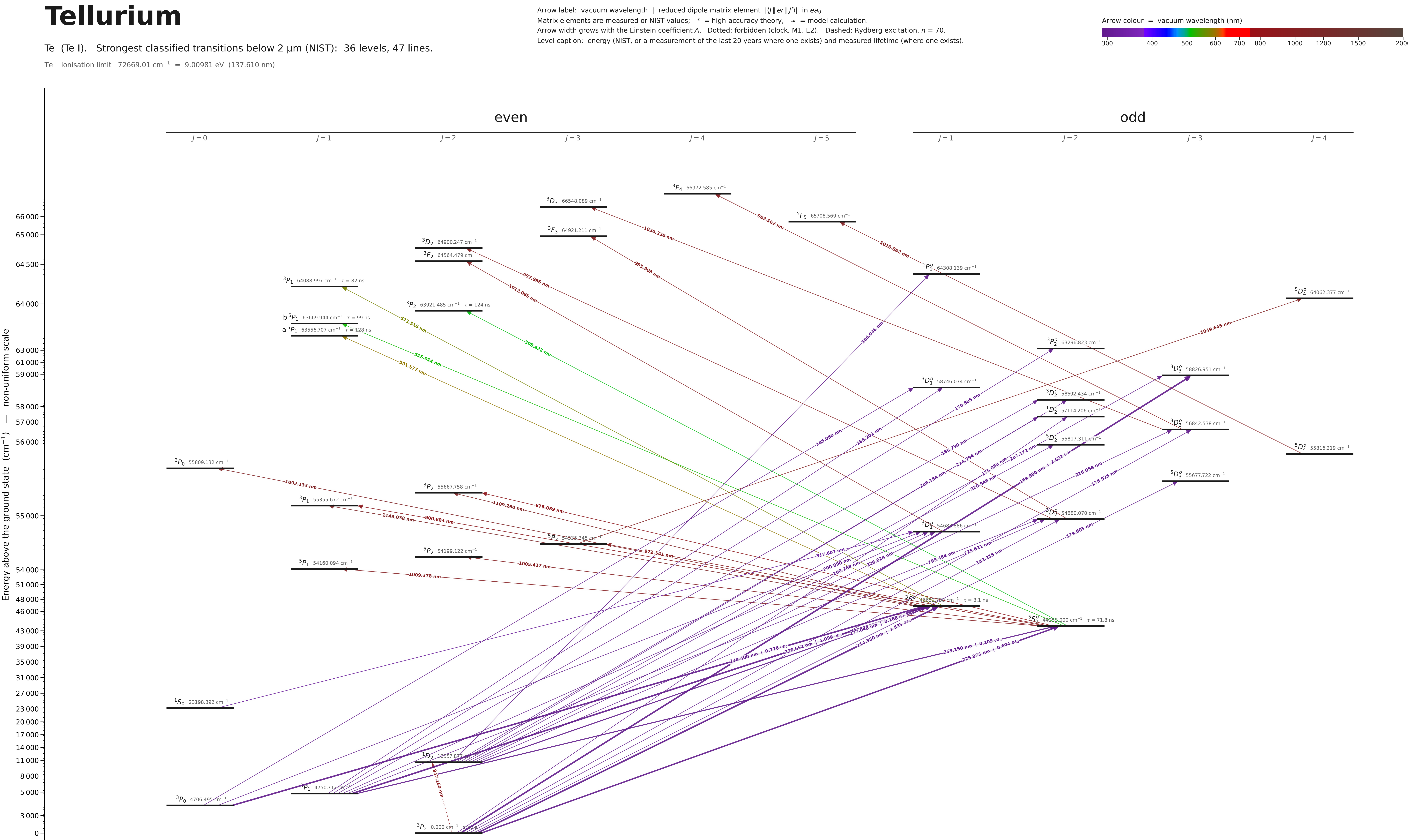




Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ P ₂ – ³ S ₂ ^o	214.3497	214.2822	1398.6139	1.835	2.310e+08	51.34	71.6	first step of the ISOLDE RILIS resonance-ionization schem...
³ S ₂ ^o – ³ P ₁	573.5175	573.3585	522.7259	–	–	1.941	–	second step of the ISOLDE RIUS resonance-ionization sche...
³ S ₁ ^o – a ⁵ P ₁	591.5770	591.4132	506.7682	–	–	1.243	–	second resonant step

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹²⁵Te hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
³ P ₂	-998(14)	–	F=3/2: +1497.00 F=5/2: -998.00
⁵ S ₂ ^o	1010.4(13)	–	F=3/2: -1515.60 F=5/2: +1010.40
³ S ₁ ^o	943(20)	–	F=1/2: -943.00 F=3/2: +471.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
³ P ₂ – ³ S ₁ ^o	214.350	120-130	-1029(11)
³ P ₂ – ³ S ₁ ^o	214.350	122-130	-829(13)
³ P ₂ – ³ S ₁ ^o	214.350	124-130	-625(13)
³ P ₂ – ³ S ₁ ^o	214.350	126-130	-426(11)
³ P ₂ – ³ S ₁ ^o	214.350	128-130	-211(11)
³ P ₂ – ³ S ₁ ^o	214.350	132-130	214(15)
³ P ₂ – ³ S ₁ ^o	214.350	134-130	402(15)
³ P ₂ – ³ S ₁ ^o	214.350	136-130	1242(21)
³ P ₂ – ³ S ₁ ^o	214.350	123-130	-815(15)
³ P ₂ – ³ S ₁ ^o	214.350	125-130	-594(19)

Shift = ν(first isotope) – ν(second isotope).